Solving & Interpreting Differential Equations

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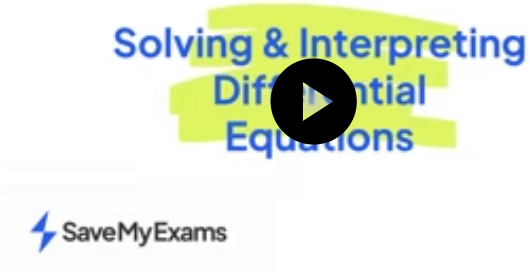

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Solving & interpreting differential equations

How do I solve a differential equation from a given model?

- Solving differential equations from a given model using the same techniques as before, for example separation of variables
- The precise integration method will depend on the type of question (see Decision Making)
- **Particular solutions** are usually required to Differential Equations
  - An **initial/boundary condition** is needed

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

$$\frac{1}{P}\,dP = -0.1\,dt$$

SEPARATE VARIABLES

$$\int \frac{1}{P}\,dP = \int -0.1\,dt$$

INTEGRATE

$$\ln|P| = -0.1t + c$$

AT $t = 0$, $P = 2$

$$\therefore\ c = \ln 2$$

USE THE INITIAL CONDITION TO FIND c

$$P = e^{-0.1t + \ln 2}$$

$$P = e^{-0.1t} \times e^{\ln 2}$$

$$P = 2e^{-0.1t}$$

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- Solutions can be rewritten in a format relevant to the model

IN THE LAST EXAMPLE, AFTER INTEGRATING THE SOLUTION WAS

$$\ln|P| = -kt + c$$

GENERAL SOLUTION

$$e^{\ln|P|} = e^{-kt + c}$$

$$P = e^{-kt} \times e^{c}$$

$e^{c} = A$

$$\therefore\ P = Ae^{-kt}$$

AT $t = 0$, $P = A$, SO A IS THE INITIAL VALUE OF P

RECOGNISING THIS TYPE OF SOLUTION CAN SAVE TIME

$$P = Ae^{-kt}$$

INITIAL VALUE

NEGATIVE EXPONENTIAL – DECAY

THE EQUATION IS IN A FORM THAT RELATES TO THE MODEL

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- The solution can be used to make predictions at other times
  - Temperature after four minutes
  - Volume of sales after another three months

How can I use the solution of a differential equation to answer modelling questions?

- Questions may ask you to interpret your solutions in the context of the problem

e.g. THE AMOUNT OF DRUG, N mg, IN A PATIENT'S BODY t HOURS AFTER INJECTION, IS MODELLED BY THE DIFFERENT EQUATION

$$\frac{dN}{dt} = -\frac{1}{8}N(t+1)$$

a) GIVEN THE INITIAL DOSE OF THE DRUG IS 30 mg FIND THE AMOUNT OF DRUG IN A PATIENT'S BODY AFTER 8 HOURS.

b) IT IS SAFE FOR A PATIENT TO HAVE ANOTHER DOSE OF THE DRUG ONCE THE AMOUNT LEFT IN THE BODY FALLS BELOW 0.1 mg. FIND, TO THE NEAREST MINUTE, HOW LONG A PATIENT WOULD HAVE TO WAIT FOR A SECOND INJECTION OF THE DRUG.

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a) $\frac{1}{N}dN = -\frac{1}{8}(t+1)dt$ VARIABLES

$\int \frac{1}{N}dN = \int\left(-\frac{1}{8}t - \frac{1}{8}\right)dt$ INTEGRATE

$\ln|N| = -\frac{1}{16}t^2 - \frac{1}{8}t + c$

$N = e^{-\frac{1}{16}t^2 - \frac{1}{8}t + c}$

$N = Ae^{-\frac{1}{16}t(t+2)}$ RECOGNISE THIS FORM

AT $t = 0$, $N = A$, $\therefore A = 30$

$N = 30e^{-\frac{1}{16}t(t+2)}$ A IS INITIAL VALUE

AT $t = 8$, $N = 30e^{-\frac{1}{2}\cdot 10}$ SUBSTITUTE VALUE OF t IN

$N = 0.202$ mg (3 sf)

b) REQUIRE $N < 0.1$

$30e^{-\frac{1}{16}t(t+2)} < 0.1$

$e^{-\frac{1}{16}t(t+2)} < \frac{1}{300}$

$-\frac{1}{16}t(t+2) < -\ln 300$

$\times -16$

$t^2 + 2t > 16\ln 300$

- IN BOTH SIDES
- $\ln\frac{1}{300} = \ln 300^{-1}$
  $= -\ln 300$

$t^2 + 2t - 16\ln 300 > 0$

$\cancel{t < -40.605...}$ $t > 8.605...$ USE CALCULATOR TO SOLVE

$t \geq 0$ IMPLIED AS TIME

8.605 HOURS IS
8 HOURS (60×0.605) MINUTES

$\therefore$ PATIENT NEEDS TO WAIT (AT LEAST)
8 HOURS 37 MINUTES
FOR THEIR SECOND DOSE

CALCULATOR SAYS 36.3 MINUTES USING 0.605 BUT t NEEDS TO BE GREATER THAN THIS

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- There could be links to other areas of A level maths – such as mechanics

THE ACCELERATION, a ms^{-2}, OF A PARTICLE TRAVELLING IN A VACUUM IS PROPORTIONAL TO HALF OF ITS VELOCITY, v ms^{-1}, AT TIME t SECONDS. GIVEN THE INITIAL VELOCITY OF THE PARTICLE IS 3 ms^{-1} AND IT TAKES 4 SECONDS FOR ITS SPEED TO DOUBLE, FIND v IN TERMS OF t AND FIND THE VELOCITY AFTER 12 SECONDS.

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· AT $t=0$, $v=3$ AT $t=4$, $v=6$

USING A MAY HIDE THE FACT THIS IS A DIFFERENTIAL EQUATION

$$a = \frac{dv}{dt} \propto \frac{1}{2}v$$

$$\frac{dv}{dt} = \frac{1}{2}kv \quad \text{WHERE } k \text{ IS A CONSTANT}$$

$$\int \frac{1}{v}\,dv = \frac{1}{2}k\,dt$$

SEPARATE VARIABLES AND INTEGRATE

$$\ln|v| = \frac{1}{2}kt + c$$

$$v = Ae^{\frac{1}{2}kt}$$

AT $t=0$, $v = A = 3$

$$\therefore\ v = 3e^{\frac{1}{2}kt}$$

USE GIVEN CONDITIONS TO FIND UNKNOWNS

AT $t=4$, $v=6$

$$\therefore\ e^{2k} = 2$$

$$2k = \ln 2$$

$$k = \frac{1}{2}\ln 2$$

$$\boxed{k = \ln\sqrt{2}}$$

$$\therefore\ v = 3e^{\frac{1}{4}t\ln 2}\ \text{ms}^{-1} \quad (*)$$

AFTER 12 SECONDS

$$v = 3e^{3\ln 2}$$

$$v = 3e^{\ln 8}$$

$$v = 3 \times 8$$

$$v = 24\ \text{ms}^{-1}$$

$$(*)\quad v = 3e^{\ln 2^{\frac{1}{4}t}}$$

$$v = 3 \times 2^{\frac{1}{4}t}$$

THIS MODEL IS REALLY BASED ON POWERS OF 2

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- Sometimes **multiple** rates of change may be involved in a model or problem
  - See Connected Rates of Change

e.g. AN AIR BUBBLE'S VOLUME IS INCREASING AT THE RATE OF $0.5\ \text{m}^3\text{s}^{-1}$. THE INITIAL RADIUS OF THE BUBBLE IS 0.1m. ASSUMING THE BUBBLE IS SPHERICAL, FIND THE RATE AT WHICH THE RADIUS OF THE BUBBLE IS INCREASING AFTER 12 SECONDS.

$$\frac{dV}{dt} = 0.5$$

AT $t=0$, $r=0.1$

$$V = \frac{4}{3}\pi r^3$$

$$\frac{dr}{dt}$$

CONVERT THE WORDED QUESTIONS INTO MATHEMATICAL STATEMENTS

$$\frac{dr}{dt} = \frac{dr}{dV} \times \frac{dV}{dt}$$

0.5

CONNECTED RATES OF CHANGE

TRYING TO FIND

CAN FIND $\frac{dV}{dr}$

$$\frac{dV}{dr} = 4\pi r^2$$

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$$\therefore \frac{dV}{dt} = \frac{1}{4\pi r^2}$$

$$\therefore \frac{dr}{dt} = \frac{1}{4\pi r^2} \times 0.5$$

$$\frac{dr}{dt} = \frac{1}{8\pi r^2}$$

WE NEED TO FIND r AT $t = 12$ AND SUBSTITUTE THIS VALUE OF r INTO $\frac{dr}{dt}$

$$8\pi r^2 \, dr = dt$$

SEPARATE VARIABLES

$$\int 8\pi r^2 \, dr = \int dt$$

$$\frac{8\pi r^3}{3} = t + c$$

INTEGRATE

AT $t = 0$, $r = 0.1$, $c = \frac{\pi}{375}$

AT $t = 12$, $\frac{8\pi r^3}{3} = 12 + \frac{\pi}{375}$

$r = 1.13$ (3 sf)

$$\therefore \frac{dr}{dt} = \frac{1}{8\pi(1.13)^2}$$

$$\frac{dr}{dt} = 0.0313 \text{ ms}^{-1} \text{ (3 sf)}$$

THIS IS ABOUT 3cm PER SECOND

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How can I interpret a model using the solution to the differential equation?

- Models may not always be realistic in the long term
  - A population will not grow indefinitely – it will reach a natural limit
  - You will be expected to interpret and comment on the model

$$\frac{dP}{dt} = \frac{1}{10}P$$

a) IF THE INITIAL POPULATION IS 600 APES, USE THE MODEL TO ESTIMATE THE POPULATION OF APES 5 YEARS LATER.

b) WHAT IS ONE LIMITATION OF THE MODEL IN THE LONG TERM? HOW CAN THE MODEL BE IMPROVED?

a) $\int \frac{1}{P}\,dP = \int \frac{1}{10}\,dt$

$\ln|P| = \frac{1}{10}t + c$

$P = Ae^{\frac{1}{10}t}$

AT $t = 0$, $P = A = 600$

$P = 600e^{\frac{1}{10}t}$

AT $t = 5$, $P = 600e^{0.5}$

$P = 989$ (3 sf)

b) THE MODEL SUGGESTS THE POPULATION OF APES WILL INCREASE INDEFINITELY FOREVER – THERE IS NO UPPER LIMIT.

AN IMPROVEMENT TO THE MODEL WOULD BE TO INTRODUCE AN UPPER LIMIT BASED ON THE SIZE/RESOURCES OF THE AREA BEING STUDIED.

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